

SAT Unit 1.5

Linear Inequalities in One or Two Variables

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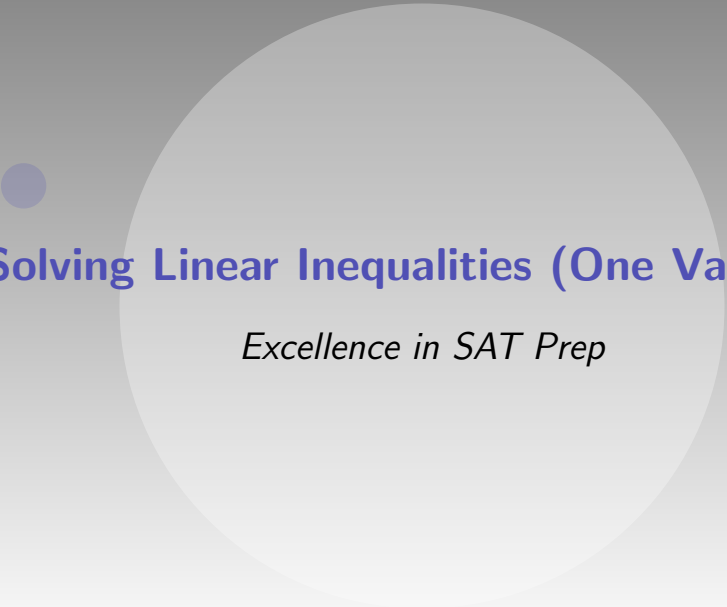
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Overview

- 1 Solving Linear Inequalities (One Variable)
- 2 Linear Inequalities in Two Variables
- 3 System of Linear Inequalities



Solving Linear Inequalities (One Variable)

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What is a Linear Inequality?

A linear inequality is similar to a linear equation, but instead of using $=$, it uses:

- $<$ (less than)
- $\leq$ (less than or equal to)
- $>$ (greater than)
- $\geq$ (greater than or equal to)

Examples:

$$x + 2 < 5 \quad \text{or} \quad 2x - 3 \geq 7$$

Solving Linear Inequalities (One Variable)

Steps:

- 1 Isolate the variable on one side.
- 2 Use inverse operations as with equations.
- 3 Important: **Flip the inequality sign** when multiplying or dividing by a negative number!

Example:

$$-3x + 2 \leq 8 \Rightarrow -3x \leq 6 \Rightarrow x \geq -2$$

Graphing One-Variable Inequalities

- Use an open circle for $<$ or $>$.
- Use a closed circle for $\leq$ or $\geq$.
- Shade the direction of the inequality.

Example: $x > 2$ is represented with an open circle at 2 and shading to the right.

Linear Inequalities in Two Variables

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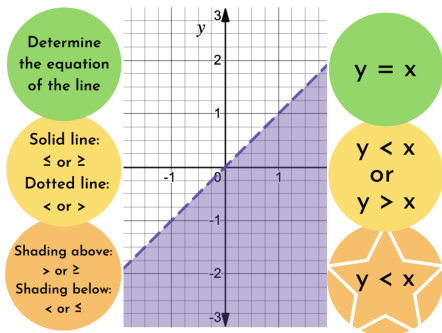
Linear Inequalities in Two Variables

A linear inequality in two variables looks like:

$$Ax + By < C \quad \text{or} \quad Ax + By \geq C$$

Steps to graph:

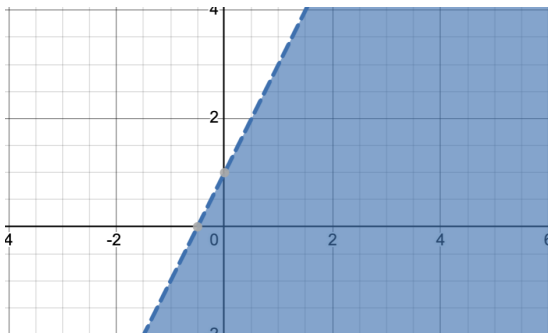
- 1 Rewrite as an equation to find the boundary line.
- 2 Use a solid line for $\leq$ or $\geq$, dashed for $<$ or $>$.
- 3 Pick a test point (usually $(0,0)$) to determine shading side.



Example: Graphing $y < 2x + 1$

- Step 1: Graph the boundary line $y = 2x + 1$ (dashed line).
- Step 2: Test $(0,0)$: $0 < 2(0) + 1$ is true.
- Step 3: Shade the region below the line.

This shaded region represents all the (x, y) pairs that make the inequality true.



System of Linear Inequalities

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System of Linear Inequalities

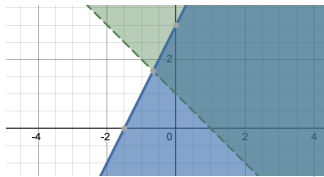
A **system of linear inequalities** is a set of two or more linear inequalities with the same variables.

Solution: The solution is the region where the shaded areas of all inequalities overlap.

Example:

$$\begin{cases} y \leq 2x + 3 \\ y > -x + 1 \end{cases}$$

- Step 1: Graph each inequality on the same coordinate plane.
- Step 2: Use dashed or solid lines depending on the symbols ($<$ or $\leq$).
- Step 3: Shade the region for each inequality.
- Step 4: The overlapping shaded region is the solution set.



Question: System of Inequalities

Given the system of inequalities:

$$y \leq x + 7$$

$$y \geq -2x - 1$$

Which point (x, y) is a solution to the system?

(A) $(-14, 0)$

(B) $(0, -14)$

(C) $(0, 14)$

(D) $(14, 0)$

Solution — Inequality System

Check each point to see which satisfies *both* inequalities:

$$y \leq x + 7 \quad \text{and} \quad y \geq -2x - 1$$

- **(A)** $x = -14, y = 0$:

$$0 \leq -14 + 7 = -7 \quad (\text{False}) \Rightarrow \text{Reject}$$

- **(B)** $x = 0, y = -14$:

$$-14 \leq 0 + 7 = 7 \quad (\text{True})$$

$$-14 \geq -2(0) - 1 = -1 \quad (\text{False}) \Rightarrow \text{Reject}$$

- **(C)** $x = 0, y = 14$:

$$14 \leq 0 + 7 = 7 \quad (\text{False}) \Rightarrow \text{Reject}$$

- **(D)** $x = 14, y = 0$:

$$0 \leq 14 + 7 = 21 \quad (\text{True})$$

$$0 \geq -2(14) - 1 = -29 \quad (\text{True}) \Rightarrow \text{Accept}$$

Answer: D

Question: Inequality Table Matching

Given the inequality:

$$y < 6x + 2$$

For which of the following tables are **all** (x, y) values solutions to the inequality?

A.

x	y
3	20
5	32
7	44

C.

x	y
3	16
5	28
7	40

B.

x	y
3	16
5	36
7	40

D.

x	y
3	24
5	36
7	48

Solution — Checking the Tables

The inequality is:

$$y < 6x + 2$$

- For $x = 3$, $6x + 2 = 20$
- For $x = 5$, $6x + 2 = 32$
- For $x = 7$, $6x + 2 = 44$

Compare each y with the threshold values:

- **A:** $y = 20, 32, 44 \rightarrow$ Not less than $\rightarrow$
- **B:** $y = 16 < 20, 36 > 32$
- **C:** $y = 16 < 20, 28 < 32, 40 < 44 \rightarrow$
- **D:** $y = 24 > 20$

Answer: C

Real-World Application

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Real-World Application

Problem: A theater has a maximum capacity of 200 people. Tickets cost \$10 for adults and \$5 for students.

Write an inequality that shows all possible combinations of adults (a) and students (s) that can attend.

$$10a + 5s \leq 200$$

This shows combinations that keep the total cost at or under \$200.

Question: Shipment Inequality Problem

A laundry service is buying detergent and fabric softener from its supplier. The supplier will deliver no more than 300 pounds in a shipment. Each container of detergent weighs 7.35 pounds. Each container of fabric softener weighs 6.2 pounds. The service wants to buy at least **twice as many** containers of detergent as containers of fabric softener. Let d represent the number of detergent containers, s the number of softener containers. Which system of inequalities best represents this situation?

A.
$$\begin{cases} 7.35d + 6.2s \leq 300 \\ d \geq 2s \end{cases}$$

B.
$$\begin{cases} 7.35d + 6.2s \leq 300 \\ 2d \geq s \end{cases}$$

C.
$$\begin{cases} 14.7d + 6.2s \leq 300 \\ d \geq 2s \end{cases}$$

D.
$$\begin{cases} 14.7d + 6.2s \leq 300 \\ 2d \geq s \end{cases}$$

Solution

- The total weight inequality is:

$$7.35d + 6.2s \leq 300$$

- The service wants **at least twice as many detergent containers** as softener containers:

$$d \geq 2s$$

- So the correct system is:

$$\begin{cases} 7.35d + 6.2s \leq 300 \\ d \geq 2s \end{cases}$$

- Answer: A**

Question: Psychologist Experiment: Inequality Problem

A psychologist conducted an experiment with 300 people, each choosing the most appealing image from a set of five arranged in random order.

- Of the first 150 participants, 36 chose the first picture.
- Among the remaining 150 participants, p chose the first picture.
- If more than 20% of all participants chose the first picture, which inequality best describes the possible values of p ?

Options:

- A. $p > 0.20(300 - 36)$, where $p \leq 150$
- B. $p > 0.20(300 + 36)$, where $p \leq 150$
- C. $p - 36 > 0.20(300)$, where $p \leq 150$
- D. $p + 36 > 0.20(300)$, where $p \leq 150$

Solution

Step 1: Total number of participants = 300

Step 2: Total number who chose the first picture = $36 + p$

Step 3: More than 20% chose the first picture:

$$36 + p > 0.20 \times 300 \Rightarrow p + 36 > 60 \Rightarrow p > 24$$

Therefore, the correct inequality is:

$$\boxed{p + 36 > 0.20(300)}, \quad \text{where } p \leq 150$$

Answer: D

Question: Triangle Inequality Theorem

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side.

If a triangle has side lengths of 6 and 12, which inequality represents the possible lengths, x , of the third side of the triangle?

- A. $x < 18$
- B. $x \geq 18$
- C. $6 < x < 18$
- D. $x < 6$ or $x > 18$

Solution

Triangle Inequality Theorem: The sum of the lengths of any two sides must be greater than the third side.

Let the three sides be: 6, 12, and x

Apply the triangle inequality:

$$6 + 12 > x \Rightarrow x < 18$$

$$6 + x > 12 \Rightarrow x > 6$$

$$12 + x > 6 \Rightarrow \text{Always true since } x > 0$$

Final range: $6 < x < 18$

Answer: C

Practices

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Graphing Inequality — Point Testing

Given the inequality:

$$y > 7x - 19$$

Which of the following points in the xy -plane satisfies the given inequality?

- A. $(-1, -32)$
- B. $(0, -20)$
- C. $(2, -3)$
- D. $(4, -14)$

Answer: Graphing Inequality — Point Testing

Step: Plug each point into the inequality $y > 7x - 19$.

A. $-32 > 7(-1) - 19 \Rightarrow -32 > -26$

B. $-20 > 7(0) - 19 \Rightarrow -20 > -19$

C. $-3 > 7(2) - 19 \Rightarrow -3 > -5$

D. $-14 > 7(4) - 19 \Rightarrow -14 > 9$

Correct Answer:

System of Inequalities — Graphical Solution

Consider the following system of inequalities:

$$y < -4x - 1$$

$$y > x + 2$$

Which point (x, y) is a solution to this system in the xy -plane?

- A. $(1, 4)$
- B. $(-1, 1)$
- C. $(-2, 2)$
- D. $(-3, -3)$

Answer: System of Inequalities — Graphical Solution

Test each point in both inequalities:

A. $(1, 4)$:

$$y < -4x - 1 \Rightarrow 4 < -4(1) - 1 = -5$$

B. $(-1, 1)$:

$$y < -4x - 1 \Rightarrow 1 < 3 \quad , \quad y > x + 2 \Rightarrow 1 > 1$$

C. $(-2, 2)$:

$$y < -4x - 1 \Rightarrow 2 < 7 \quad , \quad y > x + 2 \Rightarrow 2 > 0$$

D. $(-3, -3)$:

$$y < -4x - 1 \Rightarrow -3 < 11 \quad , \quad y > x + 2 \Rightarrow -3 > -1$$

Correct Answer: C

Inequality: Calories Burned by Jogging

Melanie burns between 120 and 135 calories for each mile that she jogs. Which inequality represents the total number of calories, y , Melanie could burn from jogging 14 miles?

A. $120 \leq y + 14 \leq 135$

B. $120 \leq y - 14 \leq 135$

C. $120 \leq 14y \leq 135$

D. $120 \leq \frac{y}{14} \leq 135$

Answer: Calories Burned by Jogging

Melanie burns 120 to 135 calories per mile.

Let y be the total calories burned in 14 miles.

Then:

$$14 \times 120 \leq y \leq 14 \times 135$$

This simplifies to:

$$120 \leq \frac{y}{14} \leq 135$$

Correct Answer: D

Challenging System of Inequalities

Consider the system of inequalities below:

$$y \leq \frac{1}{2}x + 3$$

$$y > -2x + 1$$

Which of the following points lies within the solution region of this system?

- A. $(-3, 2)$
- B. $(2, 4)$
- C. $(4, -1)$
- D. $(0, 2)$

Answer: Graphical Inequality Intersection

Inequality 1: $y \leq \frac{1}{2}x + 3$ **Inequality 2:** $y > -2x + 1$

Test option **D:** $(0, 2)$

$$\text{Check: } 2 \leq \frac{1}{2}(0) + 3$$

$$\text{Check: } 2 \geq -2(0) + 1 \Rightarrow 2 > 1$$

Correct Answer: D

Thank You!

We appreciate your time and focus.

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